

**Roads Are
No Fun**

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Four 'C's of Design

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1 Four 'C's of Design

The four 'C's of design are concepts all designers should be aware of. They are realities of any design process which represent difficulties (or opportunities) that must be acknowledged and mitigated.

1.1 Creativity

Design typically requires generation or synthesis of new ideas. This ability is called creativity. Creativity may be a natural skill among certain individuals ("think outside the box") but it can also be a result of a structured process. Brainstorming techniques exist to help create multiple novel solutions.

1.2 Complexity

Design problems are often complex, meaning they involve multiple connected parts which may also include uncertainty or lack of understanding. Humans are adapted to operating in a three-dimensional environment which leads to a difficulty of intuitively handling more than 3 or 4 parameters at a time. This is the multi-dimensional problem (curse of dimensionality). Computers are better configured to handle multiple parameters as long as the problem can be structured analytically or if data can be taken which samples the entire parameter space adequately. This leads to the second issue of complexity which is imperfect knowledge. If enough data cannot be taken to characterize the design space, then there will be uncertainty. If models are created to analytically predict the design space, the models may be simplifications which lose validity in certain regions of the design space.

1.3 Choice

Decision making is the fundamental aspect of design; if decisions are not being made among alternative choices, then design is not taking place. A designer must choose among multiple alternatives often with imperfect evaluation of those alternatives. How the choice is made is defined by the decision making process. A structured decision making process is preferable to subjective 'gut feeling' or 'engineering judgment'. Structured decision making involves identification and ranking of criteria and evaluation of those criteria among the design alternatives.

1.4 Compromise

Design nearly always involves compromise which means making concessions on desirable traits. This is necessary because of competing objectives related to physical realities. The designer must prioritize and balance the competing objectives to define the proper concessions; they become like Goldilocks looking for 'just-right'. An example is mass versus strength and stiffness; for any given material and geometry, increasing stiffness or strength also increases mass.

References

1. George E. Dieter, Linda C. Schmidt; Engineering Design Fourth Edition; McGraw Hill; 2009.